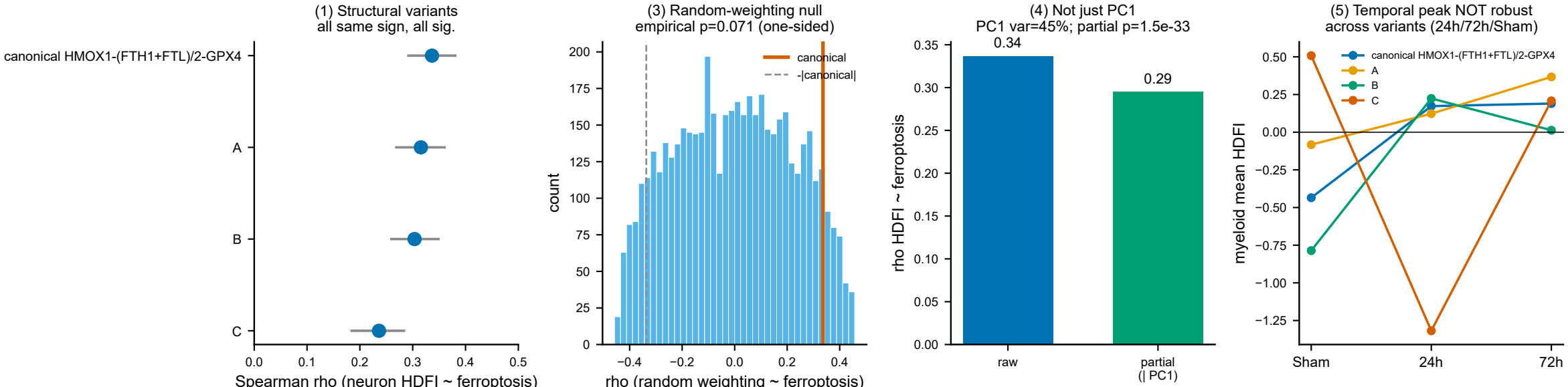
How robust is HDFI to its specification? structural-variant, leave-one-out, random-weighting-null, PCA-independence and temporal-peak tests
real mouse SAH brain atlas (GSE311852) neurons + myeloid; cell-level rho, bootstrap 95% CI (n=2000)



ersal flips, survives PC1 removal), but the headline neuronal association is largely carried by the GPX4 term SHARED with the ferroptosis outcome (leakage-free rho ~0.07); the canonical is only ~top-7% (not uniquely optimal) among random signed weightings, and the temporal peak is NOT consistent across variants